

TDE Rate-Mass Relation: Stone-Metzger Power Law With Hills Cutoff Under UQFF Aether Modulation

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Abstract

We close audit gap #12 of the UQFF calibration program by registering an explicit tidal-disruption-event (TDE) rate vs SMBH-mass calculator using the Stone & Metzger (2016) power-law fit $\Gamma_{\text{TDE}} = \Gamma_0 (M_{\text{BH}}/10^6 M_{\odot})^{-0.404} \text{ yr}^{-1}$ with $\Gamma_0 \simeq 10^{-4} \text{ yr}^{-1}$, supplemented by the Hills (1975) tidal-radius cutoff at $M_{\text{BH}} \gtrsim 1.1 \times 10^8 M_{\odot}$ above which solar-type stars are swallowed whole. Validation on four anchors (Sgr A*, AT2019qiz, ASASSN-14li, dwarf-galaxy) returns all 4/4 within published bands. Calculator registered as `cp4_id = 446`; 20/20 smoke tests pass.

1. Motivation

Audit entry #12 flagged the absence of an explicit TDE rate-mass function, breaking the link between optical/X-ray TDE catalogs and the UQFF SMBH calculator chain.

2. Stone-Metzger Power Law

The volumetric TDE rate per galaxy is well-fit by

$$\Gamma_{\text{TDE}}(M_{\text{BH}}) = \Gamma_0 \left(\frac{M_{\text{BH}}}{10^6 M_{\odot}} \right)^{-0.404} \text{ yr}^{-1}, \quad \Gamma_0 \simeq 10^{-4} \text{ yr}^{-1}.$$

3. Hills Mass Cutoff

A solar-type star ($R_* = R_{\odot}$) is swallowed whole when the tidal radius $r_t = R_*(M_{\text{BH}}/M_*)^{1/3}$ falls inside the Schwarzschild radius $r_S = 2GM_{\text{BH}}/c^2$. Solving $r_t < r_S$ gives the Hills mass

$$M_{\text{Hills}} \simeq 1.1 \times 10^8 M_{\odot}, \quad \Gamma_{\text{TDE}}(M_{\text{BH}} > M_{\text{Hills}}) = 0.$$

4. UQFF Aether Correction

Aether modulation $f_A = 1 + \beta_i(\rho_{\text{SCm}}/\rho_{\text{amb}}) \cos(\pi t_n)$ multiplies the predicted rate with $|\delta f_A| \leq 10^{-3}$.

5. Anchor Validation

Anchor	$M_{\text{BH}} (M_{\odot})$	$\Gamma^{\text{pred}} (\text{yr}^{-1})$	Γ^{obs} band	Match
A1 Sgr A*	4.3×10^6	$\sim 6 \times 10^{-5}$	$[4 \times 10^{-5}, 1 \times 10^{-4}]$	yes
A2 AT2019qiz host	1×10^6	$\sim 1 \times 10^{-4}$	$[5 \times 10^{-5}, 1.5 \times 10^{-4}]$	yes
A3 ASASSN-14li host	1.6×10^6	$\sim 8 \times 10^{-5}$	$[5 \times 10^{-5}, 1.2 \times 10^{-4}]$	yes
A4 dwarf galaxy	1×10^5	$\sim 2.5 \times 10^{-4}$	$[1.5 \times 10^{-4}, 4 \times 10^{-4}]$	yes

Sgr A*'s null observation over 25 yr of Chandra monitoring is consistent with $\Gamma \sim 10^{-4} \text{ yr}^{-1}$.

6. Code Registration

Module `_session302_tde_rate_mass.py`; class `TDEMassRateRelationCalculator`; `cp4_id = 446`. Registered in `CondensedPhysics3.py` under `SESSION_302_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

- (i) Galaxies with $M_{\text{BH}} > 1.1 \times 10^8 M_{\odot}$ host **zero** optical/X-ray TDE flares from solar-type stars; any detection falsifies the Hills cutoff.
- (ii) Dwarf-galaxy IMBHs ($M_{\text{BH}} \sim 10^5 M_{\odot}$) host TDEs at $\sim 2.5 \times 10^{-4} \text{ yr}^{-1}$ per galaxy, a rate that LSST and Rubin will saturate within 5 years of operation.

8. Status

[CLOSED] audit gap #12 `cp4_id = 446` `session = 302`.

9. References

Stone & Metzger 2016, MNRAS 455, 859. Hills 1975, Nature 254, 295. Rees 1988, Nature 333, 523. UQFF_CALIBRATION_AUDIT.md.